हिन्दुस्तान पेट्रोलियम
HP

इंजीनियर्स इंडिया लिमिटेड (भारत सरकार का उपक्रम)
ENGINEERS INDIA LIMITED
(A Govt. of India Undertaking)

Residue Upgradation Facility (RUF)
EPCC-3 Package for
Visakh Refinery Modernization Project (VRMP)

L&T Hydrocarbon Engineering

L&T- CHIYODA LIMITED

Title: (504-LZ-930A/B) GAD For Dock Leveler
Doc. No.: BO16-RUF-LT-504-MP-GN-NTPL(1)-13002

Rev. No.:
Page No: 1 of 4

OWNER : HINDUSTAN PETROLEUM CORPORATION LIMITED (HPCL)
PMC : ENGINEERS INDIA LIMITED, NEW DELHI (EIL)
UNIT : 504
PMC JOB NO. : B016
EQUIPMENT DESCRIPTION : DOCK LEVELER
EQUIPMENT TAG : 504-LZ-930A/B
P. R. NO. : BO16-RUF-LT-504-MP-PR-0014
VENDOR NAME : NIHVA TECHNOLOGIES PVT LTD
VENDOR DOCUMENT NO. : NIHVA_20-21_0157

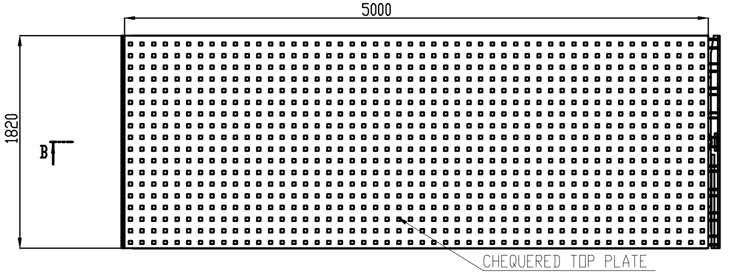
DOCUMENT SUBMISSION STATUS – LTHE/VENDOR	**REVIEW STATUS**					
	LTHE			**EIL**		
DOC. CATEGORY: (USE TICK MARK) ☑ R ☐ I	☐ Code 1 – No Comments. Proceed With Manufacture / Fabrication / Construction As Per The Document. ☑ Code 2 – Proceed With Manufacture/Fabrication/Construction As Per Commented Document. Revised Document Required. ☐ Review Code 3 – Document Does Not Conform To Basic Requirements As Marked. Resubmit For Review ☐ Code R– Document Is Retained For Records. Proceed With Manufacture / Fabrication ☐ Code V- Void (Submission Required / Not Required)			☐ Code 1 – No Comments. Proceed With Manufacture / Fabrication / Construction As Per The Document. ☐ Code 2 – Proceed With Manufacture/Fabrication/Construction As Per Commented Document. Revised Document Required. ☐ Review Code 3 – Document Does Not Conform To Basic Requirements As Marked. Resubmit For Review ☐ Code R– Document Is Retained For Records. Proceed With Manufacture / Fabrication ☐ Code V- Void (Submission Required / Not Required)		
ISSUED FOR ☐ INFORMATION ☑ REVIEW ☐ CONSTRUCTION ☐ AS BUILT	DISCIPLINE	SIGN	DATE	DISCIPLINE	SIGN	DATE

00	Issued for Information						DD-Mmm-'YY
Revision No.	**Description**	**Prepared by**	**Reviewed by**	**Approved by**	**Reviewed by**	**Approved by**	**Approved Date**

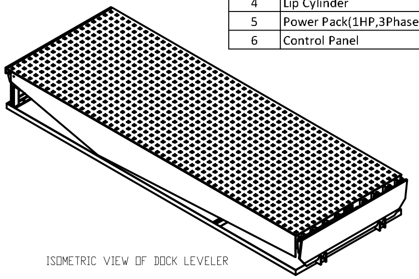
REVISION HISTORY SHEET

REV. NO.	REV. DATE	REVISED PAGE	REVISION DESCRIPTION
01	24/09/2021	NA	ISSUED FOR APPROVAL

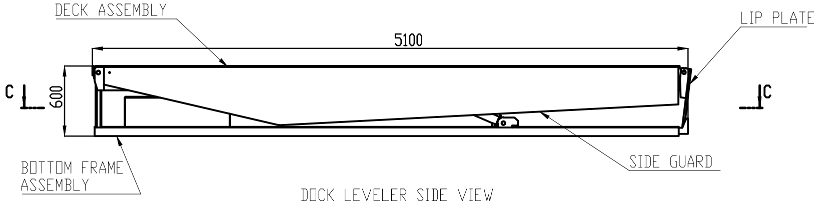
BOM(For Single Dock leveler)		
SL.NO	DESCRIPTON	QTY
1	Dock Leveler (1820 mm x 5000 mm)	1No
2	Lip (1820 x 250 mm)	1No
3	Deck Cylinder	2Nos
4	Lip Cylinder	1No
5	Power Pack(1HP,3Phase)	1No
6	Control Panel	1No



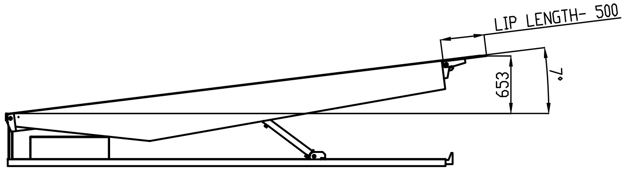
DOCK LEVELER TOP VIEW



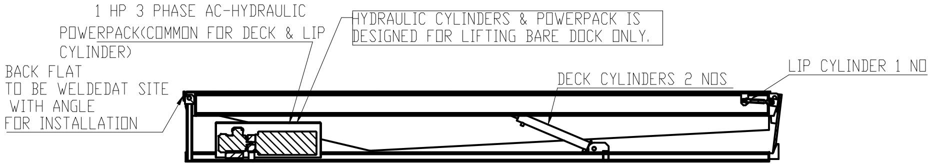
ISOMETRIC VIEW OF DOCK LEVELER



DOCK LEVELER SIDE VIEW

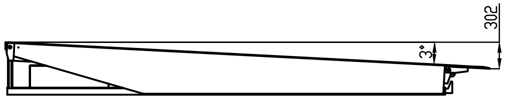


DECK UP HEIGHT - 600 MM

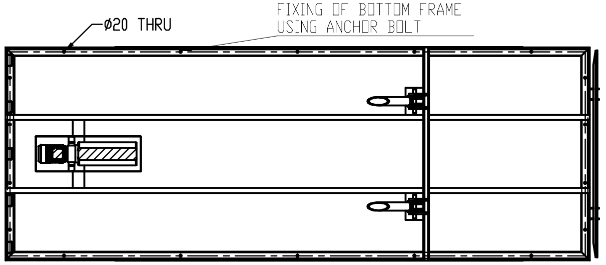


1 HP 3 PHASE AC-HYDRAULIC POWERPACK(COMMON FOR DECK & LIP CYLINDER)
HYDRAULIC CYLINDERS & POWERPACK IS DESIGNED FOR LIFTING BARE DOCK ONLY.
BACK FLAT TO BE WELDEDAT SITE WITH ANGLE FOR INSTALLATION
DECK CYLINDERS 2 NOS
LIP CYLINDER 1 NO

SECTION B-B
SCALE 1 : 15



DECK DOWN HEIGHT - 300 MM



SECTION C-C
SCALE 1 : 15

TECHNICAL SPECIFICATIONS OF PIT MOUNTED ELECRO-HYDRAULIC DOCK LEVELER -

1. SIZE OF DOCK LEVELER = 1820mm(W) X 5000mm(L)
2. OVERALL SIZE OF PRODUCT = 1820mm(W)X5100mm(L)X600mm(D)
3. CAPACITY = 6 TON POINT LOAD
4. LIP PLATE SIZE = 500mm(L)X1820mm(W)
5. DECK PLATE = CHEQUERED SHEET 1820mm(W)x5000mm(L)
6. DECK & LIP UP=DOWN OPERATION BY HYDRAULIC CYLINDER
7. DECK UP HEIGHT= (+)600 MM
8. DECK DOWN HEIGHT= (-)300MM
9. HEIGHT OF DOCK LEVELER FROM ROAD LEVEL= 900 mm,
10. OPERATION CONTROL BY PUSH BUTTONS ON CONTROL PANEL MOUNTED ON STAND NEAR DOCK LEVELER
11. ELECTRICAL SUPPLY OF EQUIPMENT= 415 VOLT 3 PHASE AC , 50HZ
12. PAINT = FIROZI BLUE
13. MATERIAL OF CONSTRUCTION = MILD STEEL AS PER IS STANDARD
14. MANUFACTURING STANDARD OF DOCK LEVELER =EN1398
15. ELECTRICAL MOTOR COMPLIANCE ZONE 1-IIC-T3 HAZARDOUS AREA .
16. PIT SIZE = 1870 mm(W)X5100mm(L)x600 mm(D)
17. DOCK LEVELER FIXING IN PIT BY ANCHOR BOLT .
18. REQUIRED PIT & ALL CIVIL WORK FOR MOUNTING OD DOCK LEVELER IS IN LTHE SCOPE .
REFER PIT DRAWING FOR CONSTRUCTION OF PIT B016-RUF-LT-504-MP-DW-NTPL(1)-13005

L&T - CHIYODA LIMITED
☐ REVIEW CODE 1 - NO COMMENTS PROCEED WITH MANUFACTURE / FABRICATION/ CONSTRUCTION AS PER THE DOCUMENT.
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☐ R - DOCUMENT IS RETAINED FOR RECORDS. PROCEED WITH MANUFACTURE / FABRICATION
☐ V-VOID

COMMENTED BY	REVIEWED BY	APPROVED BY	DATE

THIS REVIEW / APPROVAL DOES NOT RELIEVE THE VENDOR / MANUFACTURER OF THE FULL RESPONSIBILITY AS TO DESIGN CORRECTNESS OF DIMENSIONS & COMPLIANCE WITHALL APPLICABLE REGULATIONS AND WITH THE REQUIREMENTS MADE IN PURCHASER SPECIFICATIONS.

DRAWING NO.	DRAWING TITLE	REV
504-LZ-930A/B	CIVIL DRAWING OF DOCK LEVELER	

REFERENCE DOCUMENTS / DRAWINGS

DRG. CATEGORY: (USE TICK MARK) R I
SCH. DATE OF SUBMISSION
EIL REVIEW/APPROVAL CODE:
VPIL NO.:
DOCUMENT STATUS
STATEMENT OF SUB: 1st 2nd 3rd
☐ FINAL FOR CODE-1 ☑ AS BUILT
ISSUED FOR
☐ AS BUILT ☐ INFORMATION
☐ CONSTRUCTION ☐ REVIEW
SIGN: DATE:
☐ REVIEW CODE 1 - NO COMMENTS PROCEED WITH MANUFACTURE / FABRICATION/ CONSTRUCTION AS PER THE DOCUMENT.
☐ REVIEW CODE 2 - PROCEED WITH MANUFACTURE / FABRICATION / CONSTRUCTION AS PER THE COMMENTED DOCUMENT. REVISED DOCUMENT REQUIRED
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☐ V-VOID
NAME DICIPLINE SIGN. DATE

REV.	DESCRIPTION	DRN	CHECKED	APPROVED	REVIEWED	APPROVED	DATE
00	ISSUED FOR APPROVAL	SMS	VJ	AK			

OWNER: **HINDUSTAN PETROLEUM CORPORATION LIMITED**
VISAKH REFINERY MODERNIZATION PROJECT
EPCC-03 PACKAGE FOR RESIDUE UP-GRADATION FACILITY (RUF)
PMC : इंजीनियर्स इंडिया लिमिटेड ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)
LSTK CONTRACTOR: L&T Hydrocarbon Engineering
DETAILED ENGINEERING CENTER: L&T - CHIYODA LIMITED
VENDOR : NIHVA TECHNOLOGIES PVT LTD
VENDOR PO NO: REFINARY/75000-82256/VHM
VENDOR DWG. NO: NIHVA_20-21_0157
DATE 31.05.21 SCALE 1:1
TITLE: 504-LZ-930A/B GAD FOR DOCK LEVELER
DEPT: DESIGN
DRN BY: SMS
CHK BY: VJ
APP BY: AK
PROJECT: L&T-RUF
PWO NO.:
DRG. NO.: [illegible]
SH.1 OF 1
REV. 00
CAD [illegible] FILE:

DATA SHEET FOR DOCK LEVELER (504-LZ-930A/B)

Sr No	Parameter	Remark
1	Required Capacity	06 Ton
2	Leveler operation mode	AC power pack hydraulically operated
3	Lip operation mode	hydraulic
4	Operation control	by push buttons and control panel
5	Height of dock leveler from road Level	900 mm
6	Travel above & below dock level	600 mm & 300 mm
7	Platform Size (W x L)	1820 mm x 5000 mm
8	Lip size (W x L)	1820 mm x 500 mm
9	Maximum slope can be achieved	7 Degree
10	Manimum slope can be achieved	3 Degree

* Please refer Attachment-1 for standard catalog of Dock Leveler.